Abdelrahman Qasim

Artificial Intelligence Engineer

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Istanbul, Türkiye
Work-permit eligible (Turkish university graduate)

SUMMARY

AI Engineer working across LLM systems, agentic architectures, computer vision, and classical ML. Shipped a production AI pipeline at a Swiss recruitment-tech startup, deployed an LLM-powered agent end-to-end on WhatsApp, and built ML platforms from feature engineering to Dockerized APIs and live dashboards. Focused on evaluation, reliability, and contributing to production-level AI products.

PROFESSIONAL EXPERIENCE

Artificial Intelligence Engineer (Internship) — Jul 2025 - Jan 2026

ROVR — Zurich, Switzerland

- Designed and shipped an end-to-end AI processing pipeline for semi-structured recruitment data, combining deterministic NLP (rule-based parsing, scoring heuristics, pattern matching) with LLM components for structured extraction, post-processing, and output validation across diverse real-world CV layouts.
- Integrated an LLM-as-a-Judge evaluation layer with routing logic and output constraints that reduced hallucinations and ensured deterministic, production-grade behavior across batch runs of hundreds of CVs.
- Built quality-control mechanisms (feature normalization, weighted scoring, ranking stability checks) and replaced hard scoring thresholds with soft transitions — substantially reducing rank instability under input perturbations, verified via Spearman rank-correlation analysis across role profiles.
- Ran fairness and bias analysis of scoring outputs, implementing length-normalized feature caps that reduced group-level score disparities without erasing legitimate candidate differences.
- Refactored a complex AI codebase with explicit module boundaries, typed interfaces, and documented failure modes — producing engineering-grade documentation for team handoff and long-term maintenance.

EDUCATION

Bahçeşehir University | 2022-2026

B.Sc. Artificial Intelligence Engineering

Focus Areas: Machine Learning, Deep Learning, NLP, Computer Vision, Embedded Systems, MLOps, Data Science.

TECHNICAL SKILLS

Programming Languages:
Python | SQL | C++ (Embedded Systems)

Machine Learning & AI:
Supervised & Unsupervised Learning | Deep Learning | NLP | Computer Vision | Generative AI | LLM Fine-Tuning | Prompt Engineering | RAG | Embedding-Based Retrieval | LLM-as-a-Judge

Frameworks:
PyTorch | TensorFlow | Keras | Hugging Face | Transformers | Scikit-learn | OpenCV | Streamlit | Pandas | NumPy

MLOps & Deployment:
Docker | Apache Airflow | Azure ML | Google Cloud Platform | Model Deployment Pipelines | Microservices

Agents & Automation:
n8n | LangChain | OpenAI API | Qdrant | Haystack 2.x | Webhooks | Prompt Routing | Multi-Agent Orchestration

LANGUAGES

- Arabic: Native
- English: Fluent | Bilingual Proficiency
- Turkish: Intermediate (B1)

CERTIFICATES

Explainable AI **2026**

Explainable AI (XAI) Specialization

- Duke University - Developing Explainable AI
- Interpretable Machine Learning
- Explainable Machine Learning

Deep Learning **2025**

Deep Learning with PyTorch, Keras and Tensorflow Specialization

- IBM - Introduction to Deep Learning & Neural Networks with Keras
- IBM - Deep Learning with Keras and Tensorflow
- IBM - Introduction to Neural Networks and PyTorch
- IBM - Deep Learning with PyTorch
- IBM - AI Capstone Project with Deep Learning

Machine Learning Operations (MLOps) **2025**

- Duke University - MLOps Platforms: Amazon SageMaker and Azure ML
- Google Cloud - Machine Learning Operations (MLOps): Getting Started
- DeepLearning.AI - Structuring Machine Learning Projects

Generative AI **2026**

- Starweaver - GenAI Foundations and AI Agents Development

Cloud Computing **2024**

- Microsoft - Introduction to Microsoft Azure Cloud Services
- Microsoft - Azure Cloud Services
- Duke University - Cloud Virtualization, Containers and APIs
- Google Cloud - Scaling Microservices App

Machine Learning **2024**

Introduction to ML Specialization

- IBM – Exploratory Data Analysis for Machine Learning
- IBM – Supervised Machine Learning: Regression
- IBM – Supervised Machine Learning: Classification
- IBM – Unsupervised Machine Learning

Data Science **2024**

- IBM - Databases and SQL for Data Science with Python
- IBM - Data Science Methodology
- IBM - What is Data Science?
- IBM - Python for Data Science, AI & Development
- IBM - Tools for Data Science

ACADEMIC EXPERIENCE - SELECTED PROJECTS

Style-Adaptive Conversational Agent for WhatsApp — Thesis abdelqasim/whatsapp-style-agent **2025 - 2026**

• Built an LLM-powered agent fully deployed on WhatsApp, using n8n as the automation backbone for webhook handling, message routing, session management, and multi-step workflow orchestration.

• Implemented style-adaptive response generation: analyzes each user's conversational style (tone, vocabulary, formality level) and dynamically conditions LLM outputs to match — producing personalized, natural interactions at scale.

• Designed multi-turn memory management, context windowing, prompt templates, and guardrail layers to maintain coherent, safe conversations without drift.

Tech: n8n, LLMs (OpenAI API), WhatsApp Business API, Python, RAG, Qdrant, Haystack 2.x, Docker

HVAC Fault Detection & Diagnostics Platform abdelqasim/HVACC **2026**

• Built a production-grade fault detection platform for HVAC equipment using the LBNL RTU sensor dataset: sliding-window feature engineering across 6 RTU signals (mean, std, quantiles, oscillation energy, autocorrelation) feeding a LightGBM classifier with MLflow experiment tracking and model registry.

• Designed and shipped a FastAPI microservice with /predict_window,/batch_predict, and /explain endpoints supporting real-time single-window inference and bulk CSV processing; containerized the full stack (API + Streamlit dashboard + MLflow) with Docker Compose.

Tech: LightGBM, FastAPI, Streamlit, MLflow, Docker, Python, Scikit-learn, Pandas

LSTM–XGBoost BTC Trading Signal Generation abdelqasim/lstm-xgb-trading-signals **2024**

• Built a hybrid pipeline for BTC/USDT futures signal generation: LSTM learns temporal price patterns across a look-back window, its output is fed as an engineered feature into XGBoost alongside RSI, MACD, Bollinger Bands, EMA, and ATR — producing BUY/SELL/HOLD signals with a risk management engine calculating leverage, expected P&L, and risk-reward ratios. Achieved ~70–75% signal accuracy in backtesting after tuning.

• Deployed a live Streamlit dashboard with real-time Binance USDM data streaming, Plotly candlestick visualization, and signal overlay; includes dedicated hyperparameter tuning scripts for both LSTM and XGBoost.

Tech: LSTM, XGBoost, Streamlit, Plotly, ccxt (Binance USDM API), TensorFlow/Keras, Scikit-learn